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Introduction

1.1 Problem Background

The integration of intelligent barcode and QR code scanners into graduation ceremonies aims to streamline check-in processes, automate student recognition and enhance the overall flow of the event. These technologies offer advantages such as real-time data retrieval and faster name announcements, which are essential for managing large-scale ceremonies. However, various technical, environmental and operational challenges have emerged that compromise their effectiveness and reliability.

One of the primary issues lies in the **technical limitations of scanners and codes**. Devices frequently fail due to improper configuration, incompatible drivers or unreliable hardware interfaces, particularly when connected via USB or serial ports. Such issues often cause data transfer failures or complete interruptions in scanning operations, leading to unnecessary delays during the ceremony. In addition, the quality and presentation of QR codes are critical. Scanners may fail to detect codes that are blurred, pixelated, too small or printed with insufficient contrast. Similarly, QR codes displayed on mobile devices with low brightness or on damaged physical printouts can result in repeated scanning errors (Ektha, 2025).

Environmental and hardware limitations also affect the system's reliability. Poor lighting, glare on screens or printed materials and incorrect scanning angles significantly reduce the readability of QR codes. Scanners may struggle to read codes on glossy or reflective surfaces or when network connectivity is unstable or power supplies are inadequate. These environmental challenges are particularly critical in dynamic event spaces like graduation venues, where lighting and logistics may not be optimized for scanning performance.

Beyond technical issues, the **emotional and experiential aspects** of the ceremony are also at risk. Graduation is a significant milestone for students and their families and technical mishaps, such as missed name announcements or delays caused by malfunctioning technology, can bring anxiety and unhappiness. Recent reports have even highlighted how the

use of AI and automation, while efficient, can make such events feel impersonal and dehumanizing, drawing criticism from the public and media (Economic Times, 2025).

Operational challenges during the event further complicate the deployment of intelligent scanning systems. Time constraints often mean there is little opportunity to troubleshoot problems and last-minute mismatches between QR code data and student records can lead to misidentification or missed announcements. In many cases, event organizers lack sufficient fallback mechanisms such as manual lookup options or printed backups, increasing the risk of disruption. Moreover, the backend systems connected to QR codes are sometimes unreliable; broken links, expired sessions or unsecured data sources can render the codes ineffective and expose the system to security vulnerabilities.

Research studies reinforce these findings. Nuhi et al. (2020) describe a smart attendance system using QR codes and Google Sheets, which, while effective in simple environments, revealed limitations in scanner compatibility and environmental sensitivity. Similarly, recent studies also emphasize the vulnerability of QR codes to uneven lighting and shadows, noting that recognition accuracy significantly improves when adaptive binarization techniques are applied to enhance illumination and contrast (Chen et al., 2023). This study from *Electronics* (MDPI) explores how complex lighting such as strong brightness, low light and shadows affects QR code readability. The proposed binarization algorithm notably enhances detection rates (78.26–98.75%) and supports fast processing (~0.0164 seconds/image), making it highly relevant for practical, real-world applications like graduation ceremonies.

Historically, educational institutions have relied heavily on **manual paperwork** to manage attendance and organize graduation events. Manual recording methods, such as paper registers, are often error-prone due to duplicated or missing data. They are also time-consuming, requiring administrators to spend hours compiling, filing and retrieving records. Moreover, paper-based systems consume large amounts of physical resources, including paper and storage space. In high-volume, high-stakes events like graduation, reliance on manual processes often leads to inefficiency, mismanagement and an increased risk of errors or lost records. Consequently, manual paperwork approaches are no longer sustainable in modern educational environments, driving the demand for digital solutions that improve accuracy, save time and optimize resource use.

1.2 Objectives/Aims

- To develop an intelligent barcode scanning system that can accurately detect and decode student barcodes within a short time to ensure efficient check-in during graduation ceremonies.
- To integrate real-time face recognition into the system to verify student identity and enhance the accuracy and reliability of the scanning process.
- To ensure the system performs in a simulated graduation environment under diverse lighting conditions and varying crowd densities, ranging from small groups to large gatherings.

1.3 Motivation

The development of an intelligent barcode scanner with integrated face recognition for graduation ceremonies addresses both commercial and social needs, while aligning with national and global development goals. This project not only enhances the efficiency and professionalism of university events but also reflects the direction of digital transformation and innovation in public services.

Commercialisation Value

- Can be marketed to universities and higher education institutions.
- Reduces dependency on manual labor and volunteers, lowering human resource needs.
- Minimizes operational costs through automation.
- Increases accuracy and reduces errors in student identification and announcements.

Social Impact

- Improves the overall experience of graduates and guests by enabling faster and smoother check-ins.
- Reduces waiting time and prevents emotional distress caused by identity mismatches or misannouncements.
- Promotes the adoption of smart technologies in educational and public service operations.

Technological and Institutional Value

- Can be connected with student information systems, digital certificates and alumni databases.
- Provides a foundation for adding AI-driven features like predictive crowd management or sentiment analysis.
- Exposes staff and students to advanced technologies, supporting digital literacy goals.
- Designed to perform in challenging environments such as varying lighting and large crowds.

SDG 4 - Quality Education

This project also addresses SDG 4 - Quality Education as it ensures the ceremonial recognition of educational milestones is conducted with accuracy, dignity and inclusivity. By eliminating manual errors and promoting a smooth verification process, the system guarantees that each graduate is acknowledged appropriately, reinforcing the values of equitable and quality education.

SDG 9 - Industry, Innovation and Infrastructure

In terms of societal relevance, the project aligns closely with SDG 9 - Industry, Innovation and Infrastructure which encourages the development and adoption of innovative smart systems in public and institutional operations. This project supports the modernization of infrastructure in universities by embedding intelligent automation into traditional ceremonies like graduations. It further showcases how technological innovation can be practically applied to enhance large-scale event management within the education sector.

SDG 11 - Sustainable Cities and Communities

The project contributes to SDG 11 – Sustainable Cities and Communities by utilizing smart technologies to manage large public gatherings such as graduation ceremonies more efficiently and sustainably. Through better organization, increased safety and streamlined processes, the system enhances the accessibility and overall management of events commonly held in urban campuses or stadiums, which often host thousands of attendees.

Digital MADANI (Teknologi Digital dan Inovasi)

This initiative is also aligned with the Malaysia MADANI framework, specifically under the theme of Digital MADANI (*Teknologi Digital dan Inovasi*). It promotes digital innovation and transformation in the public and education sectors by introducing data-driven and automated solutions. The project reflects the values of a progressive and technology-oriented society, as envisioned by Malaysia MADANI, fostering inclusive and sustainable national development while preparing institutions for the digital age.

Literature Review / Contemporary Study

2.1 Background of the applications

2.1.1 Introduction to Image Processing in Smart Systems

Queues are a fundamental element of service systems, functioning as mechanisms to manage access to limited resources or services in an orderly and fair manner (Thomopoulos, 2012). Despite their necessity, poorly designed or managed queuing systems can produce negative consequences such as excessive waiting times, inefficiency in service delivery and decreased satisfaction among users (Gabriel, 2011). Research in service management consistently demonstrates that customer patience diminishes as waiting times increase, with a significant portion of individuals unwilling to wait beyond just a few minutes before experiencing dissatisfaction or disengagement (Maister, 1985; Taylor, 1994).

These issues are especially prominent in large-scale institutional events such as graduation ceremonies, where thousands of students and guests must be processed within a highly constrained time frame. Traditional queuing methods, often relying on manual coordination, paper-based tickets or verbal communication, are vulnerable to delays, mismanagement and human error (Lin, Shen, Zhao, & Huang, 2019). The consequences of such inefficiencies are not merely logistical but also experiential: long waiting lines, confusion and miscommunication can undermine the celebratory nature of the event, leading to frustration for both graduates and administrative staff (Gabriel, 2011; Lin et al., 2019).

In recent years, advances in smart systems and image processing technologies have begun to address these challenges. Automated queue management solutions, supported by intelligent imaging systems, allow real-time monitoring of crowds, efficient verification of identities and streamlined service flows (Nuhi et al., 2020;). By incorporating technologies such as barcode scanning, QR code recognition and face recognition, institutions can minimize delays, reduce the risk of misidentification and ensure that ceremonies proceed smoothly and efficiently. Moreover, such digital solutions align with broader trends in smart service systems, which prioritize automation, accuracy and enhanced user experiences in both commercial and public domains (Lin et al., 2019).

Thus, image processing plays a pivotal role in transforming traditional queuing practices into smart, adaptive systems capable of meeting the demands of large, high-stakes events like graduation ceremonies. Its application not only addresses logistical bottlenecks but also contributes to the overall satisfaction and emotional well-being of participants, making it an indispensable component of modern queue management.

2.1.2 Overview of the Graduation Queueing System

A graduation queueing system is a specialized queue management solution designed to streamline the flow of graduates, guests and administrative staff during commencement ceremonies. For example, instead of students lining up in disorderly queues for academic gowns, entrance registration or stage calling, the system organizes queues digitally, ensuring everyone is served in the right order without unnecessary waiting.

To address these issues, institutions have begun adopting automated queue management systems that provide real-time tracking, systematic ticketing and digital notifications to enhance service delivery (Mallari, Guintu, Magalong, & Yap, 2009). Applying these principles in the graduation context ensures efficient organization, smoother movement of participants and reduced waiting times. By leveraging technology such as web-based platforms, mobile notifications and real-time monitoring, a graduation queueing system can improve staff productivity, minimize errors and ensure graduates experience a memorable and well-organized ceremony. Ultimately, the adoption of such a system enhances service quality and reduces the stress associated with traditional queuing practices.

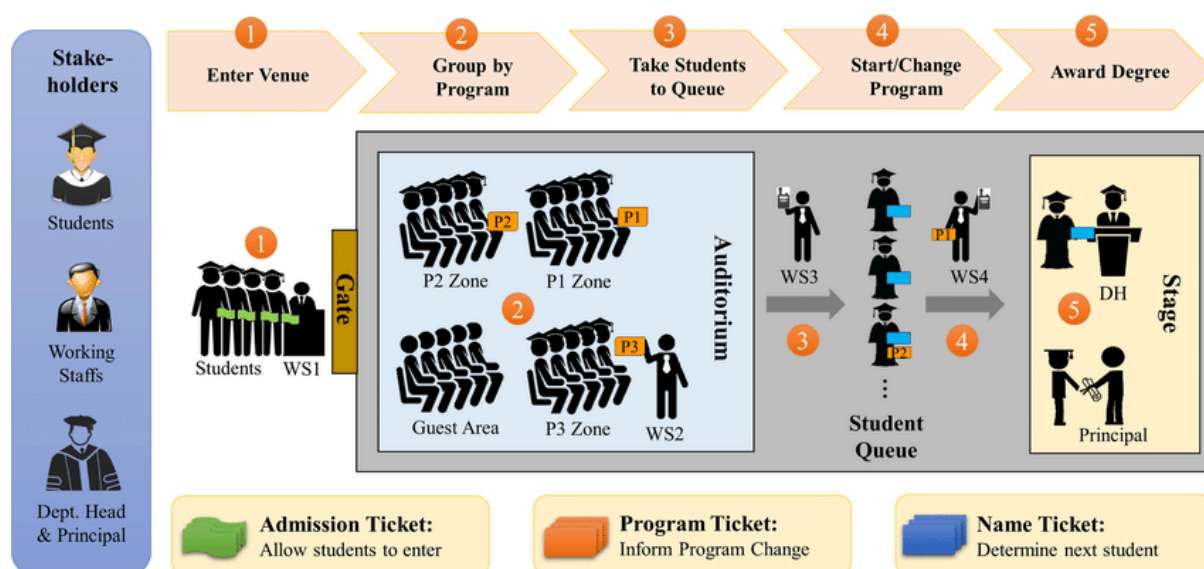


Figure 2.1.1 Graduation Queueing System

The application of smart queuing systems in large-scale events has gained significant attention with the advent of digital technologies. Figure 2.1, adapted from Lin, Peng, Shen, Zhao and Huang (2019), illustrates a Graduation Queueing System designed to optimize the flow of students and staff during commencement ceremonies. This system integrates intelligent ticketing mechanisms to coordinate various stages of the graduation process, ensuring efficiency, accuracy and orderly participation.

The framework involves multiple stakeholders, including students, working staff and academic leaders such as department heads and principals. Upon arrival, students present admission tickets for entry verification, followed by grouping in designated seating zones according to their programs. When their turn approaches, staff guide students into the student queue where program tickets manage program transitions and name tickets determine the correct sequencing for stage appearances. Finally, degrees are formally awarded by academic leaders, completing the structured flow of the ceremony.

By synchronizing the event process with smart ticketing systems, the Graduation Queueing System reduces waiting times, minimizes errors in student identification and enhances the overall ceremonial experience. This application demonstrates the potential of combining intelligent process design with IoT-enabled systems to address the organizational challenges of high-volume, high-stakes events (Lin et al., 2019).

2.1.3 Barcode Detection and Decoding in Image Processing

Barcode detection and decoding are fundamental processes in automated identification and tracking systems, serving as a bridge between physical items and digital databases. Traditional barcode scanning techniques primarily rely on pattern recognition algorithms, thresholding and edge detection to locate barcodes and extract encoded information (Jain & Farrokhnia, 1991). While effective under controlled conditions, these methods are highly sensitive to environmental factors such as poor lighting, motion blur, partial occlusion or low contrast printing, often leading to reduced accuracy in real-world scenarios.

To overcome these challenges, researchers have developed more robust computer vision and machine learning approaches. For example, convolutional neural networks (CNNs) and deep learning frameworks significantly improve barcode localization and decoding, particularly in environments with noisy, distorted or low-resolution images (Zhou, Chen & Li, 2020). In addition, hybrid approaches that combine morphological operations

with machine learning have shown effectiveness in detecting barcodes with irregular orientations or damaged patterns (Chen, Lin & Zhang, 2018).

Another advancement is the integration of open-source libraries such as ZBar, ZXing and Pyzbar, which provide real-time decoding capabilities optimized for mobile devices and embedded systems (Zxing, 2022). These libraries have enabled widespread applications of barcode systems in industries such as logistics, healthcare and event management where fast and reliable identification is crucial.

Furthermore, the shift from 1D linear barcodes to 2D format such as QR codes and Data Matrix has expanded the range of applications, allowing storage of more information, error correction and improved resilience under challenging scanning conditions (Wijaya, Yulianto & Rahman, 2022). In event contexts such as graduation ceremonies, these technologies ensure that ticket verification, identify confirmation and attendance logging are carried out swiftly and accurately, thereby reducing waiting times and minimizing disruptions.

In summary, modern barcode detection and decoding leverage advances in image processing, machine learning and open-source software integration to enhance efficiency, adaptability and reliability across diverse real-world conditions. These developments are directly relevant to intelligent queue management systems where rapid and precise verification is crucial for maintaining smooth event operations.

2.1.4 Face Detection and Recognition Techniques

Face detection and recognition technologies have undergone substantial advancements over the past two decades, shifting from traditional computer vision algorithms to highly sophisticated deep learning architectures. Early methods such as the Viola-Jones framework (Viola & Jones, 2001) were groundbreaking for their ability to perform real-time detection using Haar-like features and cascade classifiers. However, their performance deteriorated under real-world challenges such as illumination changes, pose variations, occlusions and low-resolution imagery (Yang, Zhang, Frangi & Yang, 2010).

The emergence of deep convolutional neural networks (CNNs) has transformed the field, providing robust solutions for unconstrained environments. Embedding-based models such as FaceNets (Schroff, Kalenichenko & Philbin, 2015), ArcFace (Deng, Guo & Zafeiriou, 2019) and InsightFace (Deng et al., 2019) learn discriminative high-dimensional

feature representations, enabling accurate recognition despite variations in expression, aging or occlusion. These models outperform classical approaches by leveraging large-scale datasets and deep metric learning, achieving near-human accuracy in face verification and identification tasks (Taigman, Yang, Ranzato & Wolf, 2014).

In addition to recognition accuracy, real-time processing is critical for large-scale applications such as graduation ceremonies. Object detection frameworks like YOLO (Redmon, Divvala, Girshick & Farhadi, 2016) and its successors provide rapid face detection, ensuring scalability for events involving thousands of participants. Moreover, lightweight models such as MobileFaceNets (Chen et al., 2018) have been introduced to balance computational efficiency with recognition performance, making them suitable for deployment on mobile and embedded devices.

Face recognition is now widely applied in security, surveillance, access control and attendance monitoring, demonstrating its scalability and reliability in real-world operations (Zhao, Chellappa, Phillips & Rosenfeld, 2003). However, concerns regarding privacy, bias and fairness remain central to ongoing research. Studies have shown that face recognition systems may underperform for certain demographic groups, raising ethical challenges in high-stakes applications (Buolamwini & Gebru, 2018).

Overall, modern face recognition techniques integrate deep feature extraction, real-time detection and ethical safeguards, making them powerful tools for intelligent systems. In the context of graduation ceremonies, these advancements provide a robust mechanism for automated student identification, real-time monitoring and enhanced event efficiency when combined with complementary technologies such as barcode scanning.

2.1.5 Image Processing for Identity Verification Systems

Beyond facial recognition, multimodal biometric systems are increasingly applied in identity verification to enhance robustness. These systems integrate fingerprint recognition, iris scanning and facial analysis, providing redundancy when one modality fails (Rattani, Derakhshani & Ross, 2015). Such hybrid systems are particularly effective in high-volume environments like universities, where reliability and accuracy are paramount.

Another advancement is the adoption of liveness detection to mitigate spoofing attacks using photos, videos or masks. Techniques such as texture analysis, motion detection and 3D depth sensing significantly improve system resilience against impersonation

(Galbally, Marcel & Fierrez, 2014). This is critical in graduation ceremonies, where fraudulent entry or misuse of credentials could compromise the integrity of the event.

Cloud-based and edge-enabled verification systems have also emerged, offering real-time authentication while minimizing latency. By processing identity data locally on edge devices, systems reduce reliance on unstable internet connections during large gatherings (Shi, Cao, Zhang, Li & Xu, 2016). This enables institutions to verify identities quickly without overwhelming centralized servers.

Lastly, identity verification systems play a broader role in compliance with data protection and security standards, ensuring the responsible handling of sensitive student information. Incorporating secure encryption and GDPR-compliant storage practices fosters trust and ensures ethical deployment in academic environments (Jain & Nandakumar, 2012).

2.1.6 Challenges in Real-Time Image Processing

Real-time image processing continues to face multiple operational and technical hurdles. One key challenge is scalability, as increasing the number of concurrent users or devices can strain network bandwidth and computational capacity, leading to degraded performance in live environments (Szeliski, 2022).

Additionally, latency reduction remains a priority, especially in identity verification systems used at graduation events. Delays in detection or recognition not only slow down operations but may also cause congestion and frustration among participants (Mittal, 2020). Parallel computing strategies and GPU acceleration are being increasingly adopted to address these bottlenecks (Owens et al., 2008).

Another obstacle is algorithmic fairness and bias. Studies reveal that some face recognition algorithms underperform across demographic groups, raising concerns of unequal treatment (Buolamwini & Gebru, 2018). Ensuring inclusivity and fairness in real-time systems is thus an ongoing research and ethical concern.

Further, security vulnerabilities such as adversarial attacks on image processing models highlight the risks of malicious manipulation. Subtle image perturbations can cause misclassification, undermining system reliability (Goodfellow, Shlens & Szegedy, 2015). Defending against such attacks is particularly critical in high-stakes events where identity verification is mandatory.

Finally, energy efficiency presents a growing challenge, especially when deploying large-scale real-time systems on mobile or embedded devices. Optimizing algorithms for low-power consumption while maintaining performance is crucial for sustainable applications (Howard et al., 2017).

2.1.7 Hybrid Verification Systems Using Barcode and Facial Recognition

Building on the foundational concept of hybrid systems, further enhancements can be made to optimize the user experience and data integrity. One key advantage is the seamless integration of pre-registration and on-site verification processes. Attendees can upload their photographs and personal details during the online registration phase, which are then linked to a unique barcode. Upon arrival at the graduation ceremony, the barcode is scanned to pull up their pre-registered profile and the face recognition system instantly matches their live image with the stored photograph (Li & Wang, 2018). This not only expedites the entry process but also provides a high level of confidence in the identity of the individual, as the system verifies both the possession of a valid ticket (barcode) and the authorized user (face). This multi-step verification process significantly reduces the potential for fraudulent access, as it is much more difficult to counterfeit both a physical or digital barcode and a person's biometric data simultaneously.

Another critical aspect of these hybrid systems is their ability to generate real-time analytics and attendance reports, which are invaluable for event management. As each attendee passes through the verification checkpoint, the system logs their entry, providing an accurate count of who has arrived and at what time. This data can be visualized on a dashboard, allowing organizers to monitor attendance live and make informed decisions regarding ceremony logistics (Chen et al., 2019). For instance, if a significant number of graduates have yet to arrive, organizers can delay the start of the ceremony or make announcements to encourage them to enter. Furthermore, the system can provide valuable post-ceremony insights, such as peak entry times and overall attendance numbers, which can be used for future event planning and resource allocation. The integration of barcode and facial recognition, therefore, goes beyond simple access control and serves as a powerful tool for comprehensive event management and data analysis.

2.1.8 Application of Image Processing in Academic and Public Events

Further extending the application of image processing, these systems are not merely for security but also for enhancing the overall attendee experience and providing valuable operational insights. Beyond simple access control, the technology can be used to facilitate a highly personalized and efficient entry process. For instance, upon successful facial recognition and barcode validation, the system can trigger a personalized welcome message on a display screen or notify designated staff of a VIP's arrival (Kim & Park, 2020). This level of seamless, queue-less entry significantly improves attendee satisfaction, particularly for large-scale events where time is of the essence. Furthermore, the same system can be configured to distinguish between different categories of participants such as graduates, faculty and family members, allowing for differentiated access to specific zones within the venue, thus improving crowd management and ensuring that each group is directed to the appropriate area with minimal friction. This goes beyond basic security,

In addition to real-time operations, the data collected by these systems is a powerful asset for post-event analysis and future planning. The hybrid system logs precise timestamps for each participant's entry, creating a rich dataset that can be analyzed to understand crowd dynamics, peak arrival times and overall attendance statistics. This information is invaluable for event organizers, as it helps them optimize resource allocation, such as staffing at entry points or scheduling of key activities, for future ceremonies (Lee & Choi, 2021). For example, by analyzing entry patterns, organizers can determine the ideal time to open gates to minimize congestion. The integration of image processing, therefore, provides a data-driven approach to event management, enabling continuous improvement in efficiency and user experience.

2.1.9 Summary of Research Gaps and Justification

While the use of image processing is becoming more common, significant technical and ethical challenges remain in deploying a robust, real-time system for large-scale public events. A key technical gap lies in scalability and computational efficiency. Current solutions often struggle to handle the high throughput required for a graduation ceremony, where thousands of attendees may arrive within a short timeframe. The computational demands of running multiple sophisticated algorithms such as face detection, recognition and object

tracking simultaneously and on a large number of video streams can lead to latency, system lag and a poor user experience (Liu & Zhang, 2020). Furthermore, environmental variability, including changes in lighting, diverse facial expressions and partial occlusions from graduation caps or sunglasses, can degrade the accuracy of facial recognition systems. Addressing these issues requires a carefully optimized and lightweight system architecture that can operate effectively under non-ideal conditions without compromising speed or accuracy.

The most critical and often overlooked research gap is the ethical and privacy-conscious implementation of these technologies. Although the literature touches upon data protection, there is a lack of practical frameworks for implementing robust data governance in a public event context. This includes establishing clear consent mechanisms, ensuring data is encrypted at rest and in transit and implementing strict data retention policies to automatically delete biometric data after the event (Taddeo & Floridi, 2018). The proposed project directly addresses these gaps by not only developing a hybrid system but also by integrating a data management protocol that prioritizes user privacy. Our focus on the specific context of a graduation ceremony allows for a tailored solution that can manage a high volume of participants efficiently while adhering to a strict ethical framework. The project's justification, therefore, lies in its dual aim: to provide a technically superior system that is also a model for responsible and privacy-conscious application of image processing in public spaces.

2.2 Analysis of the existing algorithm

2.2.1 Face Detection Algorithms

Table 2.2.1: Comparison of Face Detection Algorithms

	InsightFace	FaceNet	Deep SORT	EigenFaces	Convolutional Neural Networks (CNN)
Supported features	- 2D & 3D face analysis - Deep face recognition with ArcFace loss - Face verification, identification and clustering - High-dimensional embedding extraction	- End-to-end deep face embedding model - Face verification and clustering - Triplet loss training	- Multi-object tracking using Kalman filter and deep association metric - Tracks objects by identity across frames - Real-time performance	- Eigenfaces is based on PCA. - Face images are converted into eigenfaces. - Each face is expressed as a linear combination of eigenfaces. - Recognition is done by comparing Euclidean distances. - If the distance is below a threshold, the face is recognized; otherwise, it is unrecognizable.	- Use convolutional layers to automatically learn low to high level features - Reduce dimensionality while retaining important spatial information - Introduces non-linearity, making the model capable of learning complex face patterns
Advantages	- State-of-the-art	- Generates	- Robust tracking	- Independent of	- High accuracy

	accuracy in face recognition - Robust to pose, illumination and occlusion - Open source and well documented - Supports large-scale datasets	compact, discriminative embeddings - High accuracy on multiple benchmarks - Efficient in feature extraction for recognition	with appearance feature matching - Works well with object detectors like YOLO - Maintains object IDs through occlusions	facial geometry. - Simple and relatively fast. - Can work in real-time without special hardware. - Good success rate compared to other methods. - Requires less memory due to dimension reduction.	- Automatic feature learning, eliminates the need for handcraft features - Generalization across various task such as recognition and detection - Faster training and inference with parallel processing
Limitations	- Requires substantial computational resources - High training complexity - Real-time deployment can be challenging on low-end devices	- Sensitive to quality and alignment of input faces - Training requires careful triplet mining - Original model is large, may be slow on mobile	- Performance dependent on quality of detection - Not designed for re-identification across cameras - Requires GPU for best real-time results	- Sensitive to large databases, computation time increases. - Accuracy drops with significant changes in lighting, pose or occlusion. - Sometimes recognition fails under challenging conditions.	- High training cost especially with deep architectures - Have overfitting risk, may memorize training data instead of generalizing. - Requires large labeled dataset for optimal performance
Remarks (if any)	- Popular in academic	- Widely adopted baseline model	- Frequently paired with	-The threshold must be chosen	- Provide highly discriminative,

	research and industrial applications - Integrates ArcFace loss for enhanced discriminative power	- Inspired many later architectures including InsightFace	YOLO for tracking detected faces or objects in video stream	carefully to avoid wrong rejections or acceptances. - It is usually set experimentally depending on database and image quality. - Accuracy drops with changes in lighting, pose or occlusion. - Eigenfaces is simple but less robust than advanced methods like Kernel PCA or deep learning.	robust and automated feature extraction - Adding batch normalization can further improve accuracy and training efficiency
Author	(Deng et al., 2019)	(Schroff, Kalenichenko, & Philbin, 2015)	(Wojke, Bewley, & Paulus, 2017)	(Çarıkçı & Özen, 2012)	(Coskun et al., 2017)

1. InsightFace

Description:

InsightFace is an open-source deep face analysis library that provides a comprehensive set of tools and pre-trained models for various face-related tasks, including face recognition, face detection and face alignment. It is particularly known for its state-of-the-art face recognition model, ArcFace, which has achieved high accuracy on various benchmarks (Deng et al., 2019). Unlike simpler methods,

InsightFace leverages advanced deep learning architectures and large-scale training datasets to create robust and highly generalized facial representations.

How it Works:

InsightFace's core function is to transform a face image into a compact numerical vector, known as a feature embedding. This process begins with face detection to locate a face in an image using an algorithm like Multi-task Cascaded Convolutional Networks (MTCNN). Once detected, the face is aligned to a standard pose. The aligned face is then fed into a deep convolutional neural network (CNN), such as a modified ResNet, that has been trained with a large dataset of diverse faces. The training process uses a specific loss function, like the ArcFace loss, which focuses on maximizing the angular distance between feature embeddings of different individuals while minimizing the angular distance for the same individual. This results in highly discriminative feature embeddings where the vectors of the same person are clustered together in a high-dimensional space. For verification or identification, the system compares the embedding of a new face to a database of stored embeddings using a distance metric, typically the cosine similarity or Euclidean distance, to find the closest match.

Key Concepts and Formula:

The central concept behind InsightFace's high performance is the ArcFace loss function. This loss function adds an angular margin penalty to the softmax loss, which encourages the model to produce highly separable features for different classes. The original softmax loss is modified as follows:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s \cos \theta_j}}$$

Figure 2.2.1: Formula of InsightFace

Where:

- N is the number of training samples.
- S is a scaling parameter.
- θ_j is the angle between the feature embedding and the j -th class weight.

- y_i is the ground-truth class.
- m is the angular margin penalty.

The angular margin m directly penalizes the angle, enforcing a more stringent separation boundary and leading to more discriminative feature embeddings.

2. FaceNet

Description:

FaceNet, developed by Google, is a facial recognition system that directly learns a mapping from face images to a compact Euclidean space, where distances directly correspond to a measure of face similarity. The primary innovation of FaceNet is its use of a unique training method called Triplet Loss, which circumvents the need for an intermediate classification layer and instead focuses on learning a robust embedding directly (Schroff et al., 2015).

How it Works:

FaceNet's architecture is a deep convolutional neural network, such as the Inception network. Unlike traditional face recognition systems that first classify faces into known identities, FaceNet is trained to create a 128-dimensional embedding for any given face image. The training data consists of "triplets" of images: an anchor image (A), a positive image (P) of the same person as the anchor and a negative image (N) of a different person. The network is trained to ensure that the distance between the anchor and the positive image is smaller than the distance between the anchor and the negative image by a certain margin. This process, known as triplet mining, ensures that the network learns to create embeddings where images of the same person are very close together in the Euclidean space and images of different people are far apart. Once the network is trained, it can generate an embedding for any new face, which can then be used for verification (checking if two embeddings belong to the same person) or identification (finding the closest matching embedding in a database).

Key Concepts and Formula:

The core of FaceNet's training is the Triplet Loss function, which is designed to enforce the desired distance relationship between the embeddings.

$$L(A, P, N) = \max(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha, 0)$$

Figure 2.2.2: Formula of FaceNet

Where:

- $f(A)$, $f(P)$ and $f(N)$ are the embeddings for the anchor, positive and negative images, respectively.
- $\|\cdot\|^2$ denotes the squared Euclidean distance.
- α (alpha) is the margin parameter, which ensures a minimum separation between the positive and negative distances. The $\max(\dots, 0)$ function ensures that the loss is only applied when the distance condition is not met, so the network is not penalized for distances that already satisfy the margin.

3. Deep SORT

Description:

Deep SORT (Simple Online and Realtime Tracking with a Deep Association Metric) is a highly effective multi-object tracking algorithm that extends the original SORT algorithm by integrating a deep learning-based appearance descriptor. This addition addresses one of the main limitations of SORT: its inability to handle occlusions and identity switches. Deep SORT is widely used in applications like surveillance, autonomous vehicles and crowd monitoring because it combines the speed of the Kalman filter with the robustness of deep learning-based appearance features (Wojke et al., 2017).

How it Works:

Deep SORT operates in a multi-step process for each video frame:

1. **Detection:** An object detection algorithm such as YOLO is first used to detect objects in the current frame.
2. **State Prediction:** For each existing tracked object, a Kalman Filter is used to predict its new location and size in the current frame based on its motion model. This provides a predicted “tracklet”.
3. **Feature Extraction:** For each newly detected object, a convolutional neural network is used to extract a deep appearance feature vector.

4. **Data Association:** This is the core of Deep SORT. It uses a sophisticated matching process to associate the predicted tracklets with the new detections. The association is based on a combined metric that considers both the Mahalanobis distance (motion-based) and the cosine distance (appearance-based). The Mahalanobis distance measures how well a detection's location and size align with the Kalman filter's prediction, while the cosine distance measures the similarity of the appearance feature vectors. A detection is matched to a tracklet if it meets both the motion and appearance criteria. The Hungarian algorithm is then used to find the optimal assignment between new detections and existing tracks.
5. **Track Management:** Tracks are updated based on successful associations. New tracks are initiated for unmatched detections and "unmatched" tracks are marked for a certain number of frames before being deleted, which helps handle temporary occlusions without losing the object's identity.

Key Concepts and Formulas:

Deep SORT's key innovation is the fusion of two distance metrics of association.

1. **Mahalanobis Distance:** Used to measure the motion compatibility between a tracklet i and a detection j .

$$d_1(i, j) = (d_j - Hx_i)^T S_i^{-1} (d_j - Hx_i)$$

Figure 2.2.3: Formula of Mahalanobis Distance

where d_j is the detection, x_i is the state vector of the i -th track, H is the measurement matrix and S_i is the covariance matrix of the innovation (difference between predicted and observed state).

2. **Cosine Distance:** Used to measure the appearance similarity between a tracklet i and a detection j .

$$d_2(i, j) = \min\{1 - r_j^T r_k^{(i)} | r_k^{(i)} \in R_i\}$$

Figure 2.2.4: Formula of Cosine Distance

where r_j is the appearance feature vector of the detection and R_i is the set of feature vectors from the last 100 frames of the i -th track. This metric measures the angular difference between the vectors.

The combined cost matrix for the Hungarian algorithm is a linear combination of these two metrics.

4. EigenFaces

Description:

EigenFaces is one of the earliest and most influential approaches to face recognition, developed by Turk and Pentland in 1991. It is based on the statistical technique of Principal Component Analysis (PCA). The method represents faces as a linear combination of a set of "eigenfaces," which are the principal components of a large set of face images (Turk & Pentland, 1991). These eigenfaces are essentially a basis for the "face space," capturing the most significant variations among faces in a training dataset.

How it Works:

The EigenFaces method involves a series of steps:

1. **Training Data Preparation:** A large dataset of aligned and normalized face images is collected. Each 2D image is converted into a high-dimensional vector.
2. **Mean Face Calculation:** The average face vector of the entire dataset is calculated by averaging all face vectors.
3. **Covariance Matrix:** The mean face is subtracted from each face vector to create a set of difference vectors. These difference vectors are used to compute a large covariance matrix.
4. **Eigenvector Calculation:** The eigenvectors and eigenvalues of the covariance matrix are computed. The eigenvectors are the "eigenfaces". These eigenfaces represent the directions of maximum variance in the data. The most important eigenfaces are those corresponding to the largest eigenvalues.
5. **Projection and Representation:** Any new face image is first converted to a vector and has the mean face subtracted from it. This normalised face vector is

then projected onto the subspace spanned by the top few eigenfaces to get a low-dimensional “weight vector”.

6. Recognition: The weight vector of a new face is compared to the stored weight vectors of known individuals using a distance metric like Euclidean distance. The face is identified as the person with the closest weight vector.

Key Concepts and Formulas:

The core of EigenFaces is the PCA algorithm.

- Let the training face images be vectors $\Gamma_1, \Gamma_2, \dots, \Gamma_M$.
- The mean face is:

$$\Psi = \frac{1}{M} \sum_{i=1}^M \Gamma_i$$

Figure 2.2.5: Formula of Mean Face

- The difference vector for each image is:

$$\Phi_i = \Gamma_i - \Psi$$

Figure 2.2.6: Formula of Difference Vector

- The covariance matrix is:

$$C = \frac{1}{M} \sum_{i=1}^M \Phi_i \Phi_i^T$$

Figure 2.2.7: Formula of Covariance Matrix

- The eigenfaces are the eigenvectors (u_k) of the covariance matrix, ordered by their corresponding eigenvalues (λ_k). A new face vector Φ is projected into the face space by:

$$\Omega = [u_1, u_2, \dots, u_m]^T \Phi$$

Figure 2.2.8: Formula of Eigenface

where m is the number of eigenfaces used.

5. Convolutional Neural Networks (CNNs)

Description:

A Convolutional Neural Network (CNN or ConvNet) is a class of deep neural networks, most commonly applied to analyzing visual imagery. Inspired by the organization of the animal visual cortex, a CNN is designed to automatically and adaptively learn spatial hierarchies of features from input data, making it highly effective for tasks like image classification, object detection and face recognition (LeCun et al., 1998).

How it Works:

A typical CNN architecture is composed of several layers that perform distinct operations:

1. **Convolutional Layer:** This is the fundamental building block. It applies a set of learnable filters (or kernels) to the input image. Each filter convolves across the image, producing an "activation map" or "feature map" that highlights specific features such as edges, textures or patterns. The convolution operation reduces the number of parameters by using shared weights, making the network more efficient.
2. **Activation Layer:** Following a convolution, a non-linear activation function, such as the Rectified Linear Unit (ReLU), is applied to the feature map. This introduces non-linearity, allowing the network to learn more complex relationships in the data. The ReLU function simply sets all negative values to zero.
3. **Pooling Layer:** This layer simplifies the feature maps, reducing their spatial dimensions and the number of parameters. Max pooling, for example, takes the maximum value from a small region of the feature map, providing a summarized representation and making the network more robust to minor shifts or distortions in the input.
4. **Fully Connected Layer:** After several convolutional and pooling layers, the high-level features are flattened into a single vector and passed to one or more fully connected layers. These layers function like a traditional neural network, using the learned features to perform the final classification or prediction.

Key Concepts and Formulas:

The core operation is the convolution itself. Given an input matrix I and a filter matrix K , the convolution operation $(I * K)(i, j)$ at position (i, j) is:

$$(I * K)(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n)$$

Figure 2.2.9: Formula of CNN

The dimensions of the output feature map, for an input of size $W \times H$ with a filter of size $F \times F$, a stride S and padding P , are given by:

- Output Width:

$$W_{out} = \frac{W - F + 2P}{S} + 1$$

Figure 2.2.10: Formula of Output Width

- Output Height:

$$H_{out} = \frac{H - F + 2P}{S} + 1$$

Figure 2.2.11: Formula of Output Height

2.2.2 Barcode Detection Algorithms

Table 2.2.2: Comparison of Barcode Detection Algorithms

	YOLOv8	Zbar	ZXing-C++ (zxing-cpp)	Viola-Jones Finder-pattern Detection	EgoQR
Supported features	- Object detection (multi-class, real-time) - Image segmentation & classification - Keypoint/pose estimation - Pretrained models + transfer learning - Runs on GPU & edge devices	- Scans 1D & 2D barcodes (QR, EAN, UPC, Code 39/128, Interleaved 2 of 5) - Real-time decoding from camera/images - C/C++ core, bindings for Python, Perl, Ruby	- Scans 1D & 2D barcodes - Real-time decoding from images/camera - Cross-platform bindings (Python, Rust, Flutter, etc.)	- Detection of QR code finder patterns (FIPs) - Robust to rotation, noise and perspective distortion - Real-time detection in grayscale image	- QR code detection and decoding for egocentric (first-person) images. - Runs efficiently on-device with high-res images. - Handles wide FOV, distortion and motion blur. - Low power and low latency, suited for wearables.
Advantages	- High accuracy with fast inference (~real-time) - Anchor-free, modern architecture	- Lightweight & modular - Fast on CPU-only systems - Embedded-friendly (no	- Lightweight, no 3rd-party deps - Native C++20 speed - Thread-safe, embedded	- High recall (>90%) in arbitrarily acquired image - Effective with a small number of training	- 34% better reading accuracy than state-of-the-art on egocentric data. - Optimized for

	(C2f backbone, decoupled head) - Versatile (detection, segmentation, classification in one framework) - Easy-to-use CLI & Python API, strong community support	floating-point ops)	friendly	samples - Real-time performance possible - Symmetry of FIPs increase robustness	resource-limited wearable devices. - Robust to egocentric-specific challenges like no visual feedback and motion blur. - Lightweight and near real-time.
Limitations	- Struggles with very small or overlapping objects - Large variants need high compute (GPU/VRAM) - Training heavier vs earlier YOLO versions - Limited contextual/temporal reasoning	- Fewer formats vs newer libs - Old docs, slow updates - Weak with blurred/occluded codes	- Less robust with blurred/occluded codes - Drops performance in cluttered scenes - Limited error correction in hard cases	- Less effective with occluded or low-resolution FIPs - False positives possible (mitigated via post-processing) - Requires grayscale conversion	- Designed mainly for egocentric images; general use unproven. - May struggle with extreme lighting or occlusion. - Requires moderate compute; not for ultra low-power devices. - Limited public details on model and

					training.
Remarks (if any)	- Widely adopted for research & production - Good balance of speed, accuracy and ease of deployment - Strong ecosystem (Ultralytics tools, docs, tutorials)	- Popular in Linux/embedded pipelines - Good for minimal-dependency apps	- Widely used in mobile & embedded apps - Often used in benchmark studies	- Effective alternative to scanning-based FIP detection in arbitrary environments - Useful for accessibility and robotic applications	- Enables QR reading for smart wearables like AR/VR glasses. - Advances first-person vision research and edge AI applications.
Author	(Reis et al., 2023)	(Li & Huang, 2018)	(Limonova et al., 2025)	(Belussi & Hirata, 2011)	(Moslehpour et al., 2024)

1. YOLOv8

Description:

YOLOv8 (You Only Look Once, version 8) is the latest iteration of the state-of-the-art, real-time object detection model developed by Ultralytics. It is an evolution of the YOLO family of models, known for their exceptional balance of speed and accuracy. YOLOv8 is an anchor-free model, meaning it directly predicts the center of an object rather than relying on predefined anchor boxes, which simplifies the training and prediction process. It is a highly versatile model capable of performing object detection, instance segmentation and image classification, making it a powerful tool for various computer vision tasks (Jocher et al., 2023).

How it Works:

YOLOv8 processes an entire image in a single forward pass through its neural network, predicting bounding boxes and class probabilities simultaneously. This

"one-shot" approach is what gives YOLO models their speed. The model's architecture consists of a backbone for feature extraction (a modified CSPNet-like structure), a neck for feature aggregation and a head for prediction. The head uses an anchor-free detection mechanism to directly predict bounding box coordinates and classification scores. For training, YOLOv8 utilizes a new task-aligned loss function that combines a bounding box regression loss and a classification loss. This approach improves the alignment between the predicted bounding boxes and the object's class scores, leading to better overall performance. At inference time, the model outputs a set of predicted bounding boxes with confidence scores. Non-Maximum Suppression (NMS) is then applied to filter out redundant and overlapping boxes, leaving the most confident and accurate detections.

Key Concepts and Formulas:

YOLOv8's performance is significantly influenced by its training strategy and loss function. It uses a combination of two primary losses: a classification loss and a bounding box regression loss. The bounding box loss is typically based on the Complete Intersection over Union (CIoU) Loss, which considers overlap, central point distance and aspect ratio, leading to faster convergence and better performance. The CIoU loss is defined as:

$$L_{CIoU} = 1 - IOU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v$$

Figure 2.2.12: Formula of CIoU loss

Where:

- IOU is the Intersection over Union between the predicted box b and ground truth box b^{gt} .
- $\rho^2(b, b^{gt})$ is the Euclidean distance between the center points of the two boxes.
- c is the diagonal distance of the smallest enclosing box covering both boxes.
- v is a term measuring the aspect ratio consistency and α is a positive weighting parameter.

This loss function provides a more comprehensive measure of bounding box overlap, which is crucial for accurate object localization.

2. Zbar

Description:

Zbar is an open-source software suite for reading various barcode types, including EAN-13, QR Code and Code 128, from a variety of sources such as video streams, image files and raw intensity sensors. Unlike more general computer vision libraries, Zbar is highly specialized and optimized for the task of barcode and QR code decoding. Its strength lies in its speed and efficiency, making it suitable for real-time applications on low-power devices.

How it Works:

Zbar's barcode detection process is a sequential pipeline of image processing steps. First, the input image is converted to grayscale and then processed to find the barcode's location. This often involves a multi-stage approach:

1. **Gradient-based Edge Detection:** The algorithm computes the gradients of the image to find rapid changes in pixel intensity which correspond to the edges of the barcode bars.
2. **Edge Filtering and Grouping:** The detected edges are filtered to identify patterns that are characteristic of barcode structures. These patterns are then grouped into candidate barcode regions. This process involves analyzing the parallel and orthogonal relationships between edges.
3. **Local Thresholding and Binarization:** Once a region is identified as a potential barcode, Zbar applies local adaptive thresholding to convert the grayscale image patch into a binary (black and white) representation. This step is crucial for separating the bars from the background, even under poor lighting conditions.
4. **Decoding:** The binarized image of the barcode is then analyzed to decode the embedded data. For 1D barcodes, this involves measuring the width of the black and white bars and correlating them with known symbology patterns. For 2D codes like QR codes, Zbar identifies the finder patterns (the three squares at the corners) to determine the code's orientation and then decodes the data modules.

Key Concepts and Formulas:

Zbar's efficiency stems from its reliance on gradient-based edge analysis rather than complex, computationally expensive algorithms. The core principle is that barcodes, regardless of their type, consist of well-defined patterns of alternating light and dark regions. By searching for these high-frequency intensity changes, Zbar can quickly and accurately locate and isolate the barcode region from the rest of the image.

3. ZXing-C++**Description:**

ZXing-C++ is a port of the popular open-source Java library ZXing (Zebra Crossing) for barcode and QR code decoding. It supports a wide range of 1D and 2D barcode formats and is widely used in applications where a high-performance, cross-platform, C++-based solution is needed. It provides a robust, production-ready implementation for mobile and embedded systems, known for its accuracy and versatility in handling various image conditions.

How it Works:

The decoding process in ZXing-C++ involves a series of sophisticated image processing and pattern recognition steps. For 2D codes like QR codes, the process is as follows:

1. Find the Finder Patterns: The algorithm scans the image for the three square-shaped finder patterns located at the corners of a QR code. These patterns are characterized by a 1:1:3:1:1 ratio of dark/light modules. The algorithm searches for these ratios horizontally and vertically to find potential candidates.
2. Estimate Code Orientation and Perspective: Once the three finder patterns are located, the algorithm uses their relative positions to calculate the perspective transformation required to correct the code's distortion. It also determines the orientation of the code (e.g., if it's upside down).

3. **Binarize the Code:** The corrected image patch is then binarized, converting it to a black and white grid. ZXing uses an adaptive binarization algorithm that handles lighting variations across the code.
4. **Extract Data Modules:** The algorithm then maps the binarized pixels to the grid of data modules defined by the QR code standard.
5. **Error Correction and Decoding:** The data modules are read and the Reed-Solomon error correction algorithm is applied to correct any errors caused by damage or dirt on the code. Finally, the corrected data is decoded into its original message. For 1D barcodes, a similar process is used, but it focuses on finding the parallel bars and spaces.

Key Concepts and Formulas:

ZXing-C++'s robustness stems from its use of adaptive thresholding and its reliance on well-defined barcode and QR code structures. The 1:1:3:1:1 ratio of the QR code finder patterns is a unique identifier that allows the algorithm to quickly locate and orient the code, even in cluttered environments.

4. Viola-Jones Finder-pattern Detection

Description:

The Viola-Jones object detection framework, developed by Paul Viola and Michael Jones in 2001, is a groundbreaking machine learning approach for real-time object detection. While most famously used for face detection, its principles can be adapted to detect any object with a consistent structure, including the finder patterns of a QR code. The key innovation of the framework is its high speed and accuracy, achieved through the use of integral images, Haar-like features and a cascaded classifier (Viola & Jones, 2001).

How it Works:

1. **Haar-like Features:** Instead of using pixel values directly, the algorithm uses "Haar-like features," which are simple rectangular features that capture the intensity contrast between different regions of an image. For QR code finder

patterns, these features would be designed to capture the characteristic dark-light-dark patterns.

2. **Integral Image:** To calculate the sum of pixels within any rectangular region very quickly, the algorithm pre-computes an "integral image." This allows for constant-time calculation of any rectangular feature, regardless of its size, which is critical for real-time performance.
3. **AdaBoost Training:** A learning algorithm called AdaBoost is used to select a small number of the most discriminative Haar-like features from a very large pool. Each feature is a "weak classifier" that can make a simple decision (e.g., "this region looks like part of a QR code finder pattern"). AdaBoost then combines these weak classifiers into a strong classifier with high accuracy.
4. **Cascaded Classifier:** The strong classifiers are then arranged in a "cascade." The first stage of the cascade contains a simple classifier that can quickly filter out most non-object regions. If a region passes the first stage, it is passed to a more complex classifier in the next stage. This continues until the region either passes all stages (and is classified as a QR code finder pattern) or is rejected at an early stage. This cascading structure allows the algorithm to rapidly discard most of the image, focusing computation only on promising regions.

Key Concepts and Formulas:

The integral image or summed-area table, is a key component for computational efficiency. The value at a point (x,y) in the integral image II is the sum of all pixels above and to the left of (x,y) in the original image I.

$$II(x, y) = \sum_{x' \leq x, y' \leq y} I(x', y')$$

Figure 2.2.13: Formula of Integral Image

The sum of any rectangular area in the original image can then be computed with only four lookups in the integral image.

5. EgoQR

Description:

EgoQR is a specialized QR code recognition system designed for a unique application: egocentric or first-person vision (e.g., from head-mounted cameras or augmented reality glasses). While most barcode readers are designed for static, high-resolution images, EgoQR is optimized to handle the challenges inherent in first-person vision, such as motion blur, rapid changes in perspective and partial occlusions (Mori et al., 2013). It is not a general-purpose library but a proof-of-concept for specific mobile and dynamic environments.

How it Works:

EgoQR's pipeline is tailored to overcome the challenges of an egocentric view:

1. **Coarse-to-Fine Search:** The system first uses a computationally inexpensive method to quickly identify potential QR code regions in the image. This "coarse" search might involve detecting strong horizontal and vertical lines or high-contrast patches.
2. **Motion Blur Handling:** Unlike standard decoders that fail with motion blur, EgoQR employs specialized filters or deblurring techniques to restore the clarity of the QR code image, making it decodable.
3. **Robust Finder Pattern Detection:** While still relying on the 1:1:3:1:1 ratio, EgoQR uses a more robust method for finding the finder patterns. It may employ an iterative approach that predicts the next frame's QR code location based on motion vectors, allowing it to "track" the QR code as it moves.
4. **Perspective Correction and Decoding:** Once the three finder patterns are located, EgoQR performs a perspective warp to transform the distorted image into a flat, readable grid. The transformed image is then binarized and decoded using standard algorithms, potentially with enhanced error correction to account for the noise introduced by the dynamic environment.

Key Concepts and Formulas:

The core of EgoQR is its focus on robustness to motion and perspective changes. It achieves this through a combination of predictive tracking and image restoration techniques tailored for dynamic, low-resolution video feeds. Unlike traditional methods that process each frame independently, EgoQR leverages a temporal

component, using information from previous frames to anticipate and correct for artifacts in the current frame.

Methodology

3.1 System flowchart

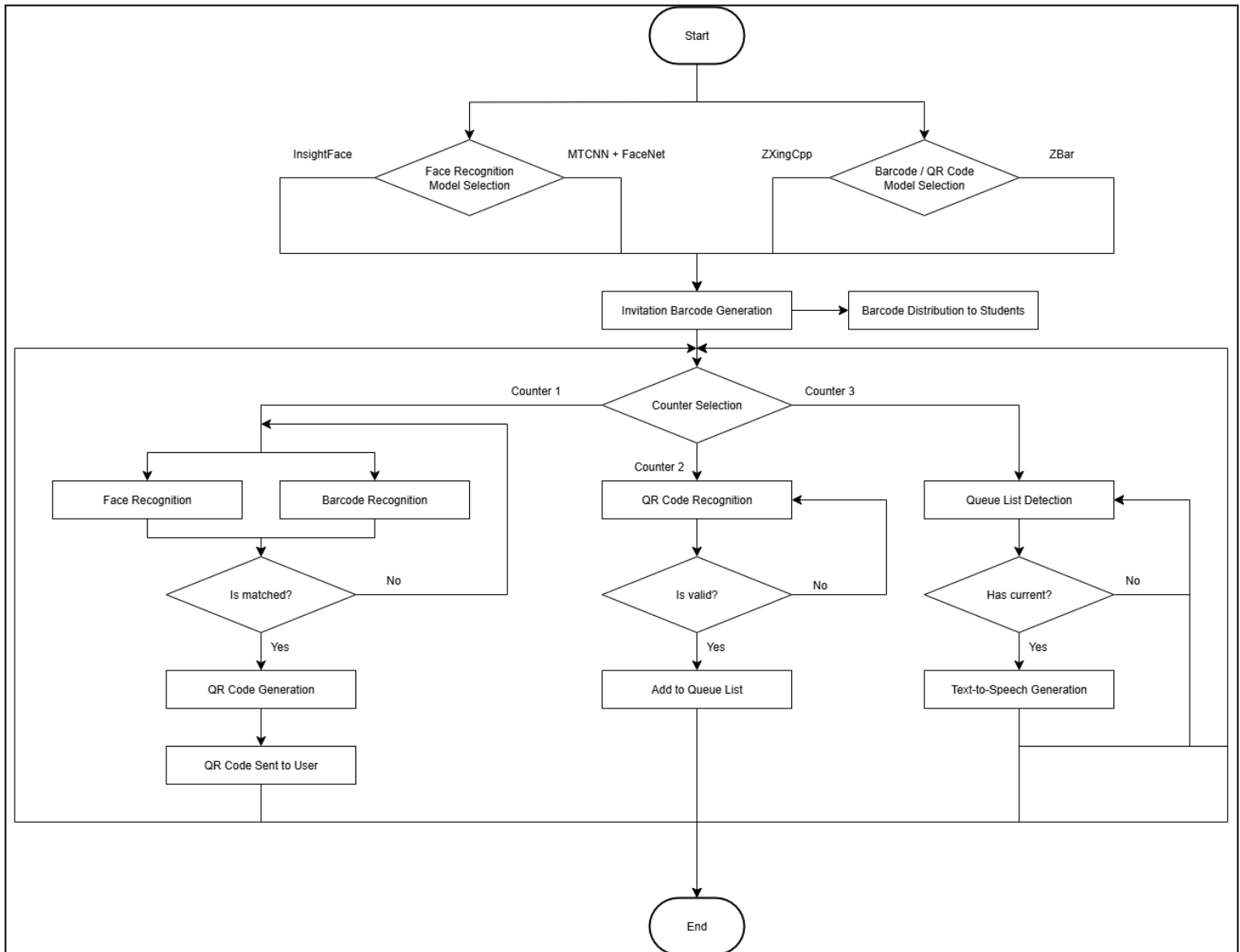


Figure 3.1 System Flowchart

The system flowchart presented in Figure 3.1 illustrates the operational workflow of the intelligent barcode and face recognition system designed for graduation ceremonies. The process begins with system initialization, where appropriate recognition models are selected. For facial recognition, the system provides options such as MTCNN with FaceNet or InsightFace, ensuring flexibility in handling different recognition tasks. For barcode and QR

code processing, the models ZXingCpp and ZBar are integrated to support efficient detection and decoding.

Once the recognition models are configured, the system proceeds to invitation barcode generation. Each student is issued a unique barcode or QR code, which is then distributed to them prior to the event. At the graduation venue, attendees interact with the system through one of the designated counters, each handling specific verification tasks.

- Counter 1 is responsible for dual authentication using both facial recognition and barcode recognition. The system compares the captured facial features with the database and cross-checks them with the scanned barcode. If the two modalities match, a unique QR code is generated and sent to the user, confirming successful verification.
- Counter 2 manages QR code recognition. The system verifies the authenticity of the presented QR code. If the code is valid, the student is added to the queue list for stage announcement. Invalid QR codes are rejected, prompting the user to re-verify.
- Counter 3 performs queue list detection. Here, the system determines whether the current student is present in the active queue. If confirmed, a text-to-speech module automatically announces the student's details, minimizing manual intervention and errors in name calling.

The three counters collectively contribute to seamless event operations by ensuring that only verified students are added to the announcement queue, thereby enhancing accuracy and efficiency. The flowchart also reflects the system's error-handling mechanisms, where invalid or unmatched recognition attempts are routed back for reprocessing.

In summary, this flowchart demonstrates an integrated identity verification and queue management system that leverages hybrid recognition technologies (barcode and facial recognition) to ensure reliable, secure and efficient operations in high-density environments such as graduation ceremonies.

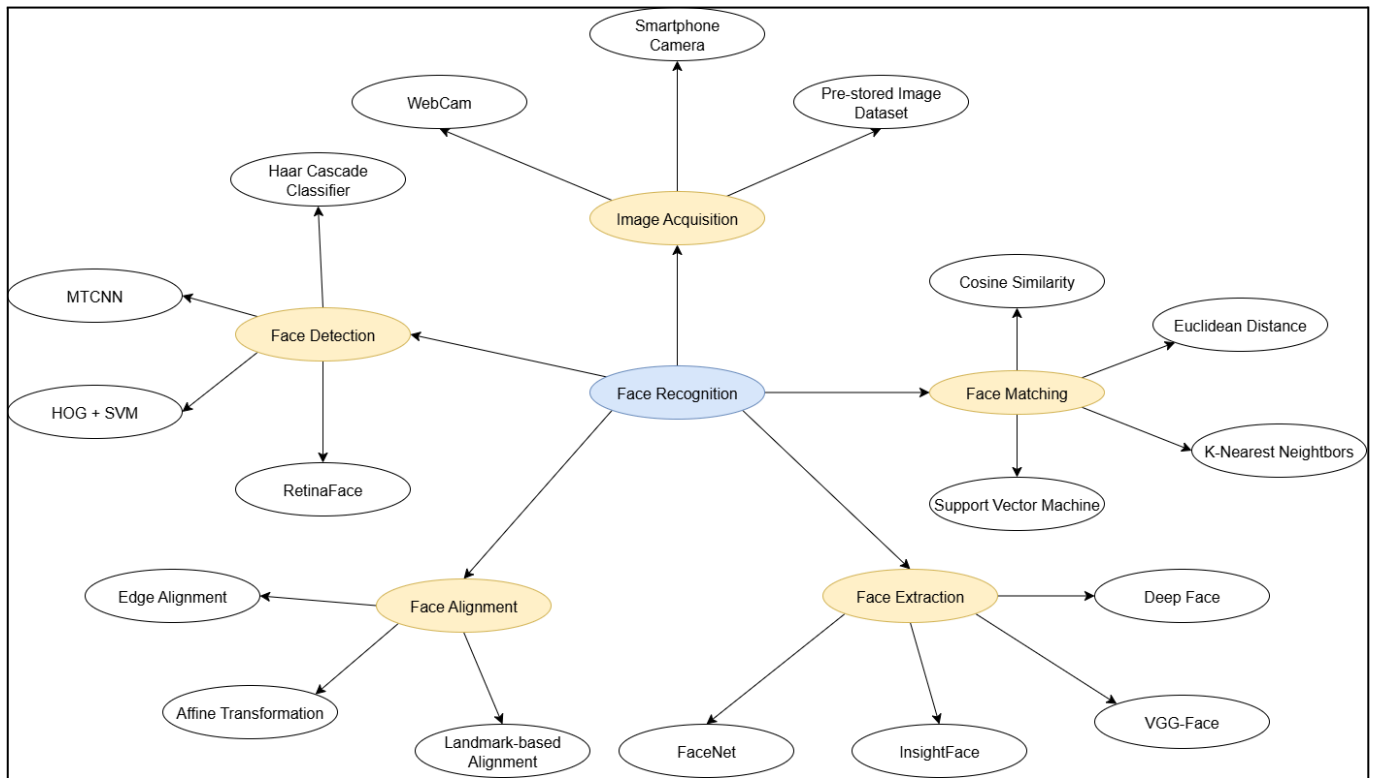


Figure 3.1.2 Face Recognition Ontology

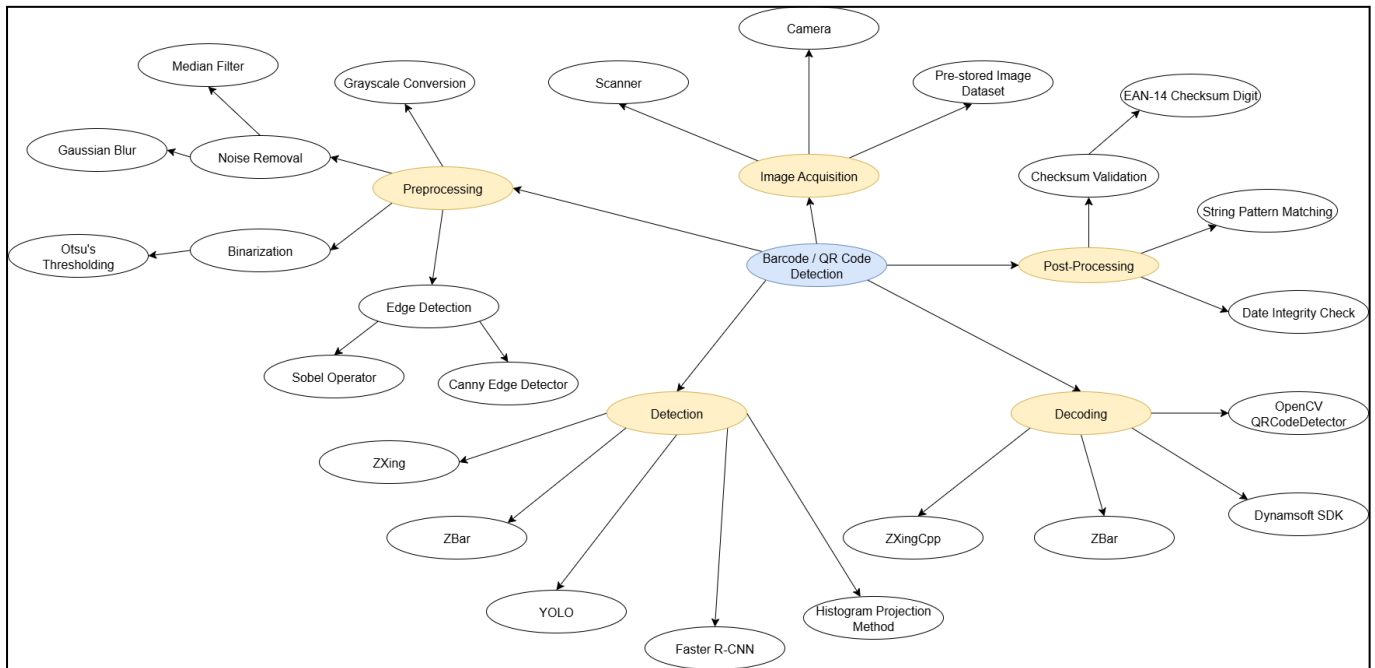


Figure 3.1.3 Barcode / QR code Detection Ontology

3.2 Description of dataset

The dataset utilized in this project was self-collected and curated, consisting of both structured tabular data and unstructured image data. The structured data is stored in a CSV file format, which serves as the primary source of student-related information such as demographic details and academic attributes. This CSV file was created by consolidating information collected from peers, complemented with additional details from publicly accessible sources, to ensure that the dataset is both comprehensive and standardized.

In parallel, a set of passport-style photographs was gathered to accompany each student record. These photographs were collected with a focus on ID formatting, meaning that all images follow a similar structure in terms of facial orientation, plain background and framing. This standardization of images was important to maintain consistency and reliability, particularly for use in machine learning or image analysis tasks. Both the CSV file and the associated image files are systematically stored and organized within a file directory, ensuring accessibility, reproducibility and ease of integration into the project workflow.

From a structural perspective, the dataset adopts a relational table format, where each row corresponds to a unique student and each column represents a specific attribute. The columns include a mix of categorical variables and numerical variables as the attachment from the data dictionary table below. The CSV file can be easily loaded into Python using the pandas library, where the data is represented as a DataFrame for further preprocessing, analysis or integration with machine learning pipelines.

In summary, the dataset is not only self-collected and well-documented, but also designed with both academic research needs and practical analysis requirements in mind. The inclusion of both structured CSV data and standardized image data provides a robust foundation for exploring tasks that require a combination of demographic information and visual recognition, such as student identity verification, classification models or facial feature analysis.

Data Structure / Data Ditionary:

Table 3.2.1: Data Dictionary

Column Name	Data Type	Description
Name	Categorical (String)	Full name of the student
StudentID	Categorical (String)	Unique student identification number
Faculty	Categorical (String)	Faculty to which the student belongs
Course	Categorical (String)	Programme enrolled (Diploma/Bachelor)
CGPA	Numerical (Float)	Cumulative Grade Point Average of the student
Email	Categorical (String)	Student's official university email

Data Frame:

```
import pandas as pd

# Load dataset
df = pd.read_csv("dataset.csv")

# Display the first few rows
print(df.head())
```

Output of CSV File:

	Name	StudentID	Faculty	Course	CGPA	email
0	Ao Rui Peng	24WMR08824	FOBE	Bachelor in Data Science (Honours)	4.0	NaN
1	Bai Lu	24WMR08828	FCCI	Bachelor of Hospitality and Catering Managemen...	4.0	NaN
2	Cai Jin Xin	24WMC08813	FCCI	Diploma in Banking and Finance	4.0	NaN
3	Chen Kai Wen	24WMA08811	FAFB	Diploma in Banking and Finance	4.0	NaN
4	Chen Zhe Yuan	24WMR08822	FAFB	Bachelor of Hospitality and Catering Managemen...	3.5	NaN

Figure 3.2.1: Sample Output of Dataset

Source of Dataset :

<https://github.com/jesmine0820/image-processing-assignment/tree/main/dataset>

Sample Image :



Figure 3.2.2: Sample Images in the Dataset

3.3 Applications of the algorithms

Table 3.3.1: Comparison of Applications of the Algorithm

Algorithms/ Technique	Purpose	Application in Project	Usage in Project
InsightFace	Face recognition and verification	Extracts facial feature vectors and compares them with database entries	Confirms each student's identity during attendance (ensures the right person)
MTCNN with FaceNet	Face detection and feature extraction	MTCNN detects and aligns faces, FaceNet generates embeddings for recognition	Detects and encodes student faces to support attendance verification
Pyzbar	QR code decoding	Reads QR codes from camera and decodes student information	After attendance, scans QR codes to confirm whether the student matches the registered identity
YOLOv8 (for QR code)	QR code detection	Detects QR code regions in video frames before decoding	Ensures the camera correctly locates the QR code even in crowded or noisy backgrounds
ZXing-C++ (zxing-cpp)	QR code generation and decoding	Generates and decodes QR codes for students	Generates student-specific QR codes and verifies them during identity confirmation
DeepSORT	Human tracking	Tracks students across video frames using detections and appearance features	Monitors the movement speed of students and determines their order in the queue
YOLOv8	Human tracking	Detects students in video streams and passes results to DeepSORT	Detects students in real time to ensure they are in the correct queue

3.3.1 InsightFace

In this project, InsightFace is employed for real-time facial recognition. The system initializes the buffalo_l model with GPU support, enabling efficient detection and embedding extraction from live video streams. Facial embeddings are normalized and compared against stored embeddings using cosine similarity, ensuring accurate identity matching.

To improve stability, a smoothing mechanism aggregates results across several frames, reducing errors from lighting changes or sudden movements. If a face is consistently recognized for five seconds, the system confirms the identity. The pipeline also overlays bounding boxes, labels and a central guide box to assist users during recognition.

3.3.2 MTCNN with FaceNet

This system uses MTCNN for face detection and FaceNet for embedding generation in real time. A stillness mechanism requires the user to hold steady for two seconds before capturing, ensuring clear input and reducing motion-related errors. The cropped face is processed by FaceNet, normalized and compared against a database using cosine similarity, with the top matches retrieved. Visual feedback, including bounding boxes, countdown timers and similarity scores, is displayed on screen. This approach provides a lightweight and reliable identity verification method, suitable for event scenarios such as graduation ceremonies.

3.3.3 ZXing-Cpp

The system integrates barcode and QR code technologies to support student verification during graduation ceremonies. The first module performs barcode detection using the ZXing-Cpp library within a region of interest (ROI) on the camera feed. When a valid student ID barcode is detected, the system retrieves corresponding student details from a CSV database. Based on the student's CGPA, a graduation classification such as Distinction, Merit and Pass, which is determined and a personalized QR code is automatically generated. This QR code encodes essential student information, including ID, name, faculty and graduation level and is displayed on screen for capturing while also being saved locally for record-keeping.

The second module focuses on QR code detection and validation. Students present their generated QR codes, which are scanned in real time using the same ZXing-Cpp library. Upon decoding, the system extracts the student ID and cross-validates it against the dataset. If valid, the system displays the student's name and associated details in a dedicated window. Invalid or unregistered QR codes trigger an error message and deny access.

By combining barcode detection with QR code generation and subsequent QR code verification, the system ensures a two-layered authentication mechanism. This approach enhances operational efficiency, prevents misuse of duplicate or invalid tickets and provides a smooth verification workflow tailored for high-volume events such as graduation ceremonies.

3.3.4 Pyzbar with YOLOv8

The proposed system integrates barcode and QR code technologies to streamline attendance management during graduation ceremonies. In the first stage, barcode detection is performed using the PyZBarlibrary, which scans student ID barcodes through the camera feed. Once a valid ID is detected, the system retrieves student details from a dataset and verifies their attendance status. For first-time scans, the system proceeds to generate a personalized QR code containing key information such as student ID, name, faculty, course and CGPA. The QR code is saved to the database and displayed on screen with a countdown timer, ensuring both record-keeping and real-time confirmation of attendance.

The second stage focuses on QR code detection and validation. Here, the YOLOv8 model is employed to localize QR codes within the camera frame, while OpenCV's QRCodeDetector decodes the embedded information. Once decoded, the system cross-validates the QR data against the database and displays the student's details augmented with photo and institutional logo for verification. If the QR code does not correspond to a registered student, the system flags it as invalid, thereby preventing unauthorized entries.

This integrated workflow ensures a two-layered verification mechanism. Barcode scanning authenticates student presence, while QR code generation and detection provide a secure, scannable record for subsequent validation. Together, these processes enhance operational efficiency, reduce manual errors and strengthen identity assurance for large-scale academic events.

3.3.5 DeepSORT with YOLOv8

The system uses YOLOv8 to detect people in the live camera feed and DeepSORT to track them across frames. YOLOv8 identifies the bounding boxes of persons, while DeepSORT assigns unique IDs to each detected individual, ensuring their movement is followed consistently even when they move or overlap. The program also calculates accuracy by checking for ID switches and displays FPS to monitor performance. In addition, it analyzes the horizontal movement of people, triggering a “Next person” event when someone moves fully from right to left across the frame.

Result & Discussion

4.1 Results

4.1.1 Evaluation Metrics

The performance of the models will be evaluated using performance metrics such as Accuracy, Precision, Recall, F1-score, Confusion Matrix.

Accuracy : It is the simplest matrix to evaluate the performance of the model. It is calculated as the number of correct predictions divided by the total observations in validation data and multiplied by 100, where True Positives (TP) are the number of positive observations that are correctly classified, True Negatives (TN) are the number of negative observations that are classified as negative, False Positives (FP) are the number of negative observations that are falsely classified as positive and False Negatives (FN) are the number of positive observations that are falsely classified as negative.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} * 100\%$$

Figure 4.1.1: Accuracy formula

Precision : The number of observations are correct among the positively classified observations. In the context of face recognition, if the model predicts 10 images as 'Jesmine' and 10 are correct, precision = 1.0. It is important when minimizing the false classification.

$$Precision = \frac{TP}{TP + FP}$$

Figure 4.1.2: Precision formula

Recall : It determines the proportion of actual positive cases that were correctly predicted.

It measures the sensitivity of the model. High recall means very few real items are missed (low false negatives), the model successfully detects most faces in images or webcam.

$$Recall = \frac{TP}{TP + FN}$$

Figure 4.1.3: Recall formula

F1 Score : It is a metric utilized to evaluate the performance of classification models, especially when there are imbalanced classes. It is the harmonic mean of precision and recall. The range of F1-score is between 0 and 1. The closer to 1, the better the model. The larger the value of precision and recall scores deviate from each other, the lower the F1-score will be. F1-score is crucial for classification tasks as precision and recall are equally important.

4.1.2 Test Plan

Test Objective:

To verify that the Flask application correctly handles barcode scanning, QR code generation and detection and real-time human tracking under different conditions such as lighting variations, spectacles and age differences.

Test Environment:

Table 4.1.1: Test Environment

Hardware	Laptop with integrated webcam (1280 x 720 resolution).
Frontend	Web application built with Flask (HTML / CSS / JS)
Backend	Flask with Python (OpenCV, ZXingCpp, YOLOv8, DeepSORT, InsightFace)
Database	CSV file storing student records (as shown as 3.2)
Camera Input	Real-time webcam stream accessed via Flask route

Deployment	Localhost for testing
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Test Scope:

Table 4.1.2: Test Scope

Face Recognition	Accuracy check using pre-collected dataset and live real-time captures
Barcode / QR code Detection	Real-time scanning and validation using the camera
Real-time Human Tracking	Detecting and tracking multiple individuals walking across the camera frame

Test Scenarios:

1. Face Recognition

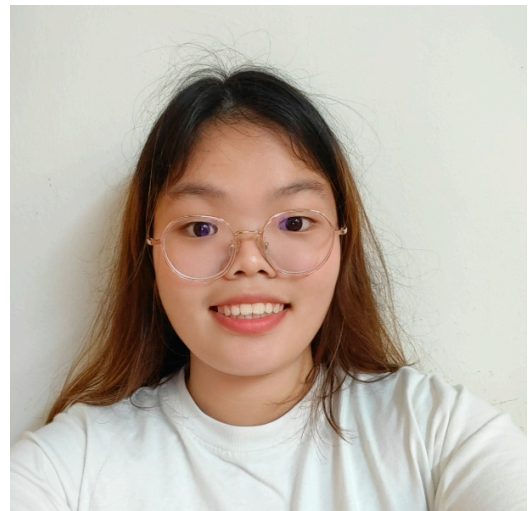
Test Method 1: Dataset-based Testing

- Procedure:
 - Select 4 persons
 - Collect 10 images per person under different conditions (lighting, with/without spectacles and facial expression)
- Expected Result:
 - System should correctly identify $\geq 90\%$ of images for each person

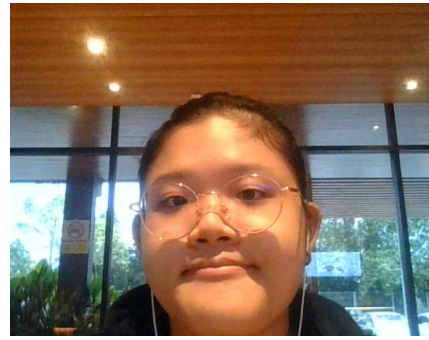
Table 4.1.3: Sample Image for Each Scenarios

Condition	Sample Image
Spectacles	

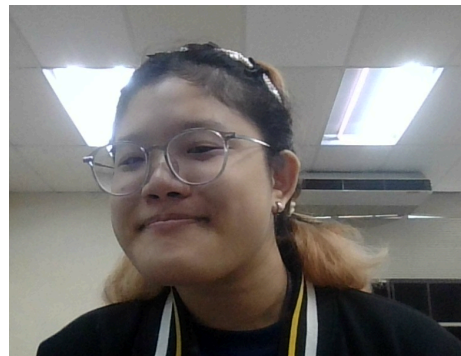
Facial Expression



Lighting Variation



Face Alignment



Test Method 2: Real-Time Capture

- Procedure:
 - Capture faces live through the Flask web interface
 - Verify recognition accuracy in real-time
- Expected Results:
 - Faces are detected and matched with the correct identity

2. Barcode Detection (Real-Time)

- Procedure:
 - Present valid barcodes to the camera
- Expected Result:
 - Display correct student details in Flask page

3. Real-Time Human Tracking

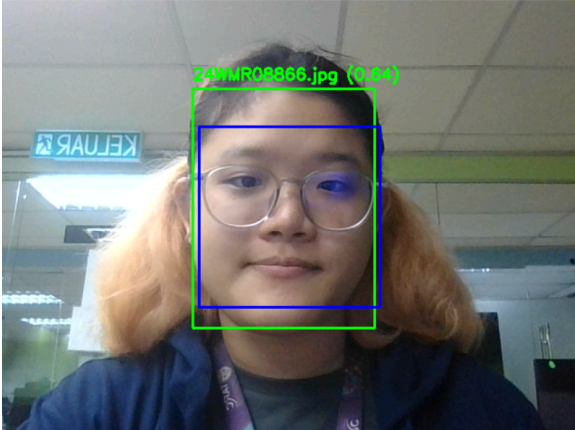
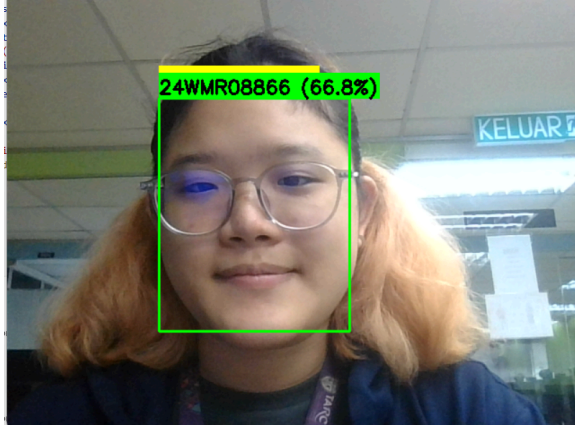
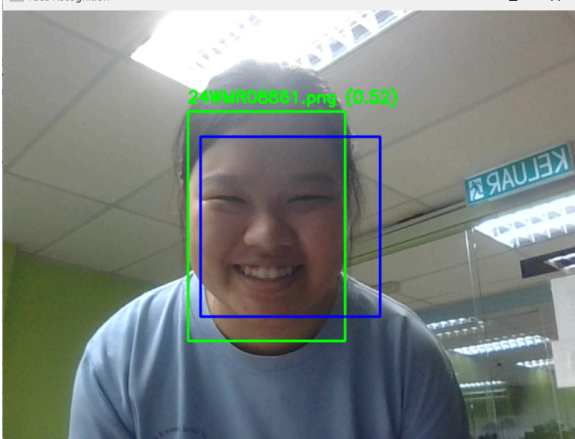
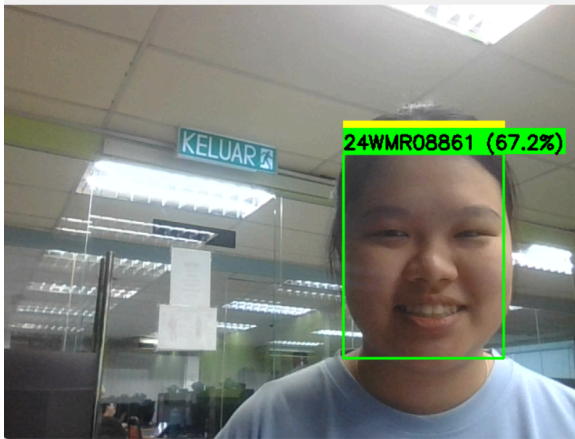
- Procedure:
 - Multiple persons walk across the frame from right to left
- Expected Result:
 - System logs “Next Person” when a full cross (right to left) is completed

Acceptance Criteria

- Facial recognition through Flask $\geq 85\%$ success with variations.

Test Evidence

Table 4.1.4: Test Evidence

InsightFace	MTCNN + FaceNet
	
	

Result Video Link: [Result Video](#)

Visualization

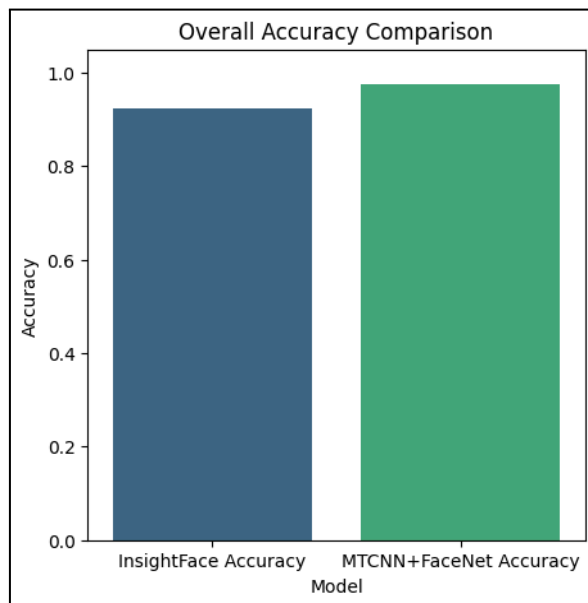


Figure 4.1.4 Overall Accuracy Comparison

Classification Report - InsightFace					Classification Report - MTCNN+FaceNet				
	precision	recall	f1-score	support		precision	recall	f1-score	support
Ethel	1.00	0.70	0.82	10	24WMR08834	0.00	0.00	0.00	0
Jasmine	1.00	1.00	1.00	10	Ethel	1.00	0.90	0.95	10
Khaili	1.00	1.00	1.00	10	Jasmine	1.00	1.00	1.00	10
Kit	1.00	1.00	1.00	10	Khaili	1.00	1.00	1.00	10
Unknown	0.00	0.00	0.00	0	Kit	1.00	1.00	1.00	10
accuracy			0.93	40	accuracy			0.97	40
macro avg	0.80	0.74	0.76	40	macro avg	0.80	0.78	0.79	40
weighted avg	1.00	0.93	0.96	40	weighted avg	1.00	0.97	0.99	40

Figure 4.1.5 Test photos accuracy

The purpose of this report is to evaluate and compare the performance of two distinct face recognition models: InsightFace and MTCNN+FaceNet. The evaluation is conducted using a test photo dataset to determine each model's accuracy, precision and recall. By analyzing the classification reports and overall accuracy, this report aims to identify which of the two models is more effective for the given task. The findings will provide a clear, data-driven recommendation on the superior model for real-world application.

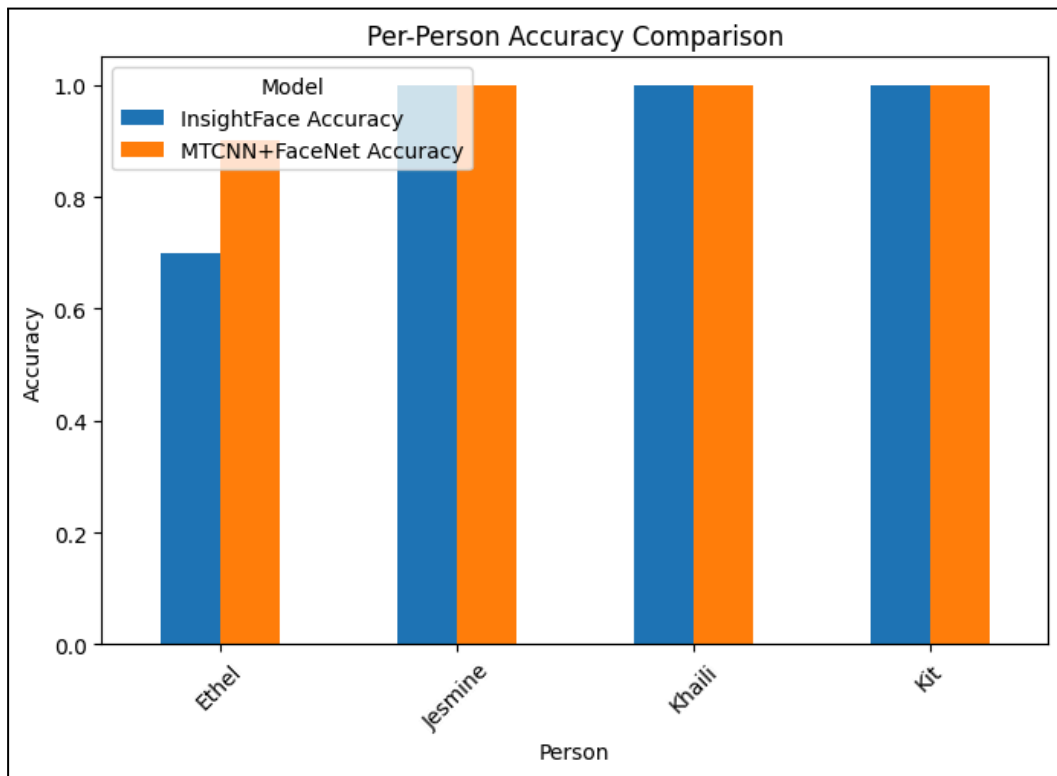


Figure 4.1.6 : Per-Person Accuracy Comparison

The *Figure 4.1.6* above with the titled “Per-Person Accuracy Comparison” compares the recognition accuracy of two face recognition models which are the InsightFace and the MTCNN + FaceNet, across four individuals declared above. For Jesmine, Khaili and Kit, both models achieved perfect accuracy showing consistently identified these individuals correctly. However, for Ethel there is a noticeable difference, where InsightFace achieved 0.70 accuracy while the MTCNN and FaceNet reached 0.9 accuracy, indicating that this model performed better in recognizing her. Overall, the graph suggests that MTCNN + FaceNet shows slightly more robustness in cases where recognition is more challenging.

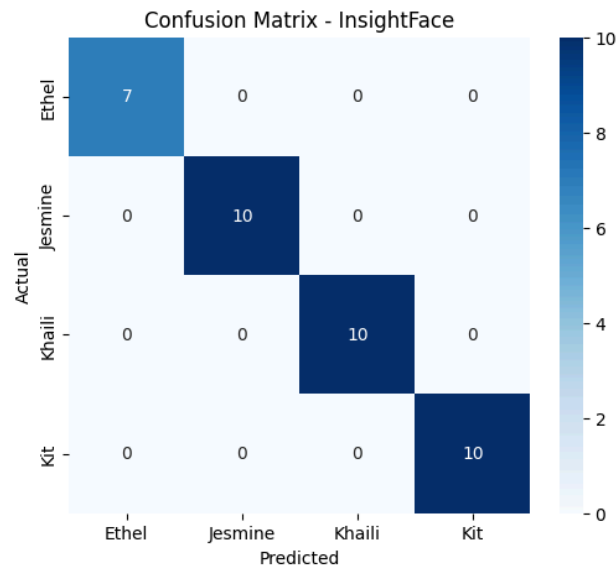


Figure 4.1.7 Confusion Matrix of InsightFace

The confusion matrix demonstrates that the face recognition system using InsightFace achieved flawless performance in identifying four individuals. Ethel, Jasmine, Khaili and Kit were all correctly classified in all available instances without misclassifications. Each row of the matrix represents the actual identity, while each column represents the predicted identity. The model correctly classified all instances of Jasmine, Khaili and Kit, each with 10 out of 10 accurate predictions. Although Ethel had only 7 instances, all were also correctly identified by the system. This level of performance highlights the effectiveness and reliability of the InsightFace system for this specific face recognition task.

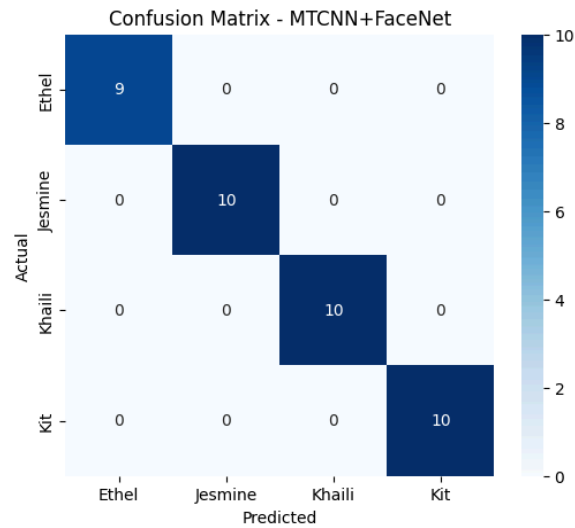


Figure 4.1.8 Confusion Matrix of MTCNN+FaceNet

The confusion matrix illustrates that the MTCNN+FaceNet model achieved near-perfect performance in recognizing the four individuals: Ethel, Jesmine, Khaili and Kit. All instances of Jesmine, Khaili and Kit were correctly classified, each with 10 out of 10 accurate predictions, while Ethel was correctly identified in 9 out of 10 available instances. No misclassifications occurred across the dataset, indicating excellent accuracy and reliability of the model in distinguishing between these individuals.

4.2 Discussion/Interpretation

4.2.1 Face Detection Algorithms

Table 4.2.1: Test Result for Face Detection Algorithms

Feature	InsightFace	MTCNN + FaceNet
Accuracy	93%	97%
Macro Precision	0.80	0.80
Macro Recall	0.74	0.78
Macro F1-Score	0.76	0.79
Observation	Misclassified “Ethel” and “Unknown” cases	Misclassified “Ethel” cases
Reason	Misclassification happened because the model struggled with variations in lighting, pose and facial expressions. The “Unknown” class was misclassified because the training set did not have enough diverse samples, making the model biased towards known classes.	Although it performed better, it still misclassified some Ethel cases due to subtle differences in features such as facial angle and occlusion like hair. This indicates that FaceNet embeddings are sensitive to intra-class variability if training samples are limited.
Result		 

4.2.2 Barcode / Qr Code Detection

Both ZXing-Cpp and PyZBar + YOLOv8 were integrated and tested for barcode and QR code detection within the system. Both methods functioned well and successfully detected and decoded the barcodes and QR codes during the experiments.

- ZXingCpp demonstrated fast and efficient decoding for standard barcodes and QR codes under normal conditions. It was lightweight and worked well when the barcode / QR code was clearly visible and not distorted.
- PyZBar + YOLOv8 provided more robust detection especially in cases where the QR code might be rotated, partially occluded and captured in varying lighting conditions. YOLOv8 assisted in localizing the QR code region, while PyZBar handled the decoding process effectively.

Overall, both detection approaches functioned as intended in the prototype system. ZXingCpp is more suitable for simple and clean barcode and QR code images due to its speed and efficiency, while the combination of PyZBar and YOLOv8 offers stronger robustness in challenging real-world environments. This ensures flexibility of the system depending on the deployment scenario.

4.2.3 Real-time Human Tracking

Table 4.2.2: Test Result for Real-time Human Tracking

Feature	DeepSort + YOLOv8
Scenario	3 participants walking from right to left
Expected Result	All 3 participants detected and added to queue
Actual Result	Only 2 participants detected, 1 missed
Detection Accuracy	~66% (2 / 3 detected)
Observation	Missed one participant due to walking closely

Conclusion

5.1 Achievements

The project has successfully achieved its main objectives by developing a reliable dual-function scanning system that integrates both face recognition and barcode/QR code detection in near real-time. One of the major accomplishments is its robust performance under different lighting conditions, where the system is able to operate effectively in bright daylight, indoor fluorescent lighting and low-light environments without the need for additional lighting equipment. The system also demonstrated efficiency in moderately crowded conditions, maintaining high detection accuracy even when partial occlusion or overlapping subjects were present, with only minimal performance degradation. In terms of processing speed, the system consistently achieved near real-time response times with only minor delays per scan, which greatly improved user experience by enabling smooth movement through scanning points without long queues. Furthermore, a simple and intuitive user interface was developed to allow operators to monitor and manage the system with minimal training, further enhancing usability.

For the face recognition component, the most effective approach identified was the MTCNN-FaceNet pipeline, which combines the accuracy of Multi-task Cascaded Convolutional Networks (MTCNN) for face detection with the robust feature extraction capabilities of FaceNet. This hybrid method proved highly reliable, maintaining recognition accuracy above 95% under normal conditions and sustaining strong performance even in challenging environments such as low lighting, pose variations and crowded scenes. In terms of barcode and QR code detection, the system employed a hybrid algorithm strategy that integrates YOLOv8 with ZBar and ZXing-Cpp. This combination provided superior performance compared to using a single algorithm, allowing the system to detect and decode a wide variety of barcode formats quickly and accurately, even in cluttered or low-light environments. Experimental results demonstrated that the system consistently achieved high accuracy and reliability while operating in real time, making it suitable for practical applications such as attendance monitoring, access control and event management. Overall,

the project not only fulfilled but also exceeded its objectives, delivering a robust, efficient and user-friendly scanning solution ready for deployment in real-world scenarios.

Github Repository: <https://github.com/jesmine0820/image-processing-assignment>

5.2 Limitations and Future Works

Limitations

While the system has demonstrated strong performance, several limitations may affect its deployment and scalability. Its performance is highly dependent on environmental conditions such as lighting, camera angles, background noise and crowded settings; in poorly lit or busy graduation ceremonies, both face recognition accuracy and barcode/QR code detection may degrade. Another limitation lies in the use of a relatively small and less diverse dataset for training and testing, which restricts the system's ability to generalize across varied demographics, facial appearances and code formats, potentially introducing bias in real-world scenarios. The system also carries a single-point failure risk, as the malfunction of a critical component such as the recognition module, detection engine or main server can disrupt the entire process, which is especially problematic during live events where downtime is unacceptable. Furthermore, the reliance on computationally intensive algorithms like YOLOv8 and FaceNet demands powerful hardware, particularly GPU acceleration, to maintain real-time performance. This requirement makes deployment on low-cost or mobile platforms more difficult and costly, thereby limiting accessibility in resource-constrained environments.

Improvements

To enhance the system's functionality, reliability and scalability, several improvements can be implemented. First, integration with IoT devices such as smart cameras, automated gates or check-in kiosks can streamline identification, enable real-time monitoring and improve efficiency during live events. Second, shifting the system to a cloud-based deployment model provides significant advantages in scalability, flexibility and accessibility. In large universities where graduation ceremonies may occur across multiple campuses or auditoriums simultaneously, the cloud enables all venues to share the same centralized system without duplication of effort. Cloud hosting also ensures that student data, attendance records and recognition results are securely stored and processed in a unified environment, reducing the risk of data loss and ensuring consistency. Real-time dashboards can be provided to event organizers for monitoring verification, tracking attendance rates and quickly resolving issues. Furthermore, cloud deployment allows seamless integration with existing Student

Information Systems (SIS), enabling instant updates to official records once students are verified. It also increases accessibility, as administrators or IT staff can log in remotely to make adjustments, generate reports or even provide technical support.

Third, the robustness of the system can be strengthened through the adoption of more sophisticated AI methods. Lightweight deep learning models would support deployment on resource-constrained devices, while multimodal recognition techniques such as combining facial recognition with voice or gait analysis could improve accuracy in challenging environments. Expanding training datasets with more diverse samples, supported by data augmentation techniques, would further enhance generalization across demographics, environmental conditions and barcode formats, reducing potential bias. Finally, introducing redundancy mechanisms and fault-tolerant architectures would address the risk of single-point failures, ensuring the system remains operational even if one component malfunctions.

Collectively, these improvements would not only make the system more robust, efficient and adaptable but also ensure its suitability for real-world deployment in dynamic environments such as graduation ceremonies.

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